

HP INK TANK COMPETITIVE INTELLIGENCE

Thailand — Technical Implementation Brief (Version 2)

POC for David Chiu | HP vs Epson, Canon and Brother

Data period: 28 May–28 August 2026 (last 3 months / 90 days)

Purpose: This brief defines the data scope, sources, output structure and UI requirements for the POC. Please use it as the implementation guide for scraping, data preparation, RAG/query capability and the web app.

1. Project Overview

Build a competitive intelligence POC for the Ink Tank printer space in Thailand, focused on HP versus the three key competitors: Epson, Canon and Brother.

Category focus: Ink Tank printers — inkjet printers that use refillable ink tanks rather than cartridge replacements.

Audience: Consumers using printers at home / for personal use, and small businesses (under 100 employees).

Output language: English only; original Thai content can be retained as source evidence.

Data period: Last 3 months / 90 days (~28 May to 28 Aug 2026).

The output is a POC web app, not a full-scale dashboard or comprehensive data set. The objective is to demonstrate the quality and potential value of the competitive intelligence.

2. What the POC Should Demonstrate

The POC should demonstrate whether the available market, media and commerce signals can answer the key competitive questions in the brief.

It should focus on evidence/ insights that can be compared across **HP, Epson, Canon and Brother** and, where possible, trended over the last 3 months.

Online visibility: Where each brand is active across paid media (ads), social media and e-commerce platforms, and the relative level of observable activity.

Marketing and sales push: The creatives, messages, brand / product positioning, promotions and products / SKUs being pushed, including observable response signals.

Consumer response and competitive implications: What consumers are saying, the main positive / negative themes, and observable signals of competitive strengths, weaknesses, opportunities or issues for HP.

3. Technical Implementation / Scraping Approach

Tooling: Use existing Apify and Bright Data scrapers that are already available / licensed. Do not build custom scrapers from scratch unless necessary; custom development is not recommended for this POC.

Crawling scope: Target relevant brand / product pages, sellers, ads and social accounts rather than entire e-commerce platforms. **Follow the source lists and keywords in Sections 5–8.**

Evidence: Retain source URLs and underlying content / evidence wherever possible, together with the original publication / listing date and the date / time captured where relevant.

4. POC Web App (Dashboard) & RAG / Query Capability

Dashboard: Show representative sample data and insights across advertising, social and e-commerce. Do not expose the full scraped data set; prioritise a small number of high-quality examples that demonstrate value.

RAG / query capability: Build a RAG engine that allows users to ask natural-language questions across the collected data and generate evidence-backed answers, insights and charts, with answers traceable to the underlying source evidence.

Hosting and language: Host the app on Vercel, preferably with password protection. All dashboard data, labels and insights should be presented in English.

5. Ad Libraries / Advertising Intelligence Sources

Source	Use
Google Ads Transparency Center	Search brand + Thailand for relevant Ink Tank advertising activity, creatives, messaging, offers / promotions and campaign characteristics.
Meta Ad Library — Facebook + Instagram	Capture relevant HP / Epson / Canon / Brother Ink Tank ads, creative formats, messaging and promotions.
YouTube	Use where relevant to identify video advertising / creative approaches and complement Google Ads Transparency findings.

Search for HP, Epson, Canon and Brother in Thailand, focusing specifically on Ink Tank printers and the last 3 months of activity.

Capture relevant ad creatives, messaging, offers / promotions, formats and other observable campaign characteristics.

Search logic:

[Brand] + Thailand + Ink Tank (relevant product terminology)

Example: “HP” + “Thailand”, “Epson” + “Thailand”, “Canon” + “Thailand”, “Brother” + “Thailand”

6. Social Channels (Brand-Owned Social Channels)

Crawl official Thailand Facebook, Instagram, TikTok, LinkedIn and YouTube accounts for HP, Epson, Canon and Brother, focusing on Ink Tank printer-related activity during the last 3 months.

Secondary benchmarking: Where relevant, also review official regional / global brand accounts to identify broader creative themes, messaging and product positioning that may inform Thailand activity.

Prioritise **printer / Ink Tank-related content** rather than general brand content.

Brand	Facebook	Instagram	YouTube
HP	HP Thailand Facebook	HP Instagram	HP YouTube (KIV)
Canon	Canon Thailand	Canon Thailand Instagram	Canon Thailand YouTube
Epson	Epson Thailand Facebook	Epson Thailand Instagram	Epson Thailand YouTube

Brand	Facebook	Instagram	YouTube
Brother	Brother Thailand Facebook	Brother Thailand Instagram	Brother Thailand YouTube

7. E-commerce Sources

Prioritise Shopee Thailand, Lazada Thailand, TikTok Shop Thailand and JIB, with Advice and Power Buy as secondary sources if feasible.

Focus specifically on HP, Epson, Canon and Brother Ink Tank printers and the last 3 months of observable activity.

Do not crawl entire platforms; target relevant brand / product pages and sellers.

Priority	Platform	Why it matters for this POC
1	Shopee Thailand	Largest mass-marketplace channel; broad electronics / printer assortment, seller competition, pricing, promotions, ratings / reviews and sales signals.
2	Lazada Thailand	Major marketplace with strong electronics coverage, official-brand presence, pricing, promotions, reviews and seller data.
3	TikTok Shop Thailand	Important social-commerce channel; useful for promotions, creator-led selling, LIVE commerce and campaign activity.
4	JIB Thailand	Specialist IT / electronics retailer with a strong printer assortment and useful price, availability and observable sales / engagement signals.

7.1 E-commerce SKU-level Data to Capture

For each relevant Ink Tank SKU under each brand, capture the following:

Brand	Metrics / fields to capture
HP	Brand; Model / SKU; Product title; Current price (THB/฿); Original / RRP price where shown; Discount %; Promotion / voucher; Seller; Official brand store / authorised seller indicator; Stock availability; Units sold / sales indicators where visible; Rating / reviews; Number of reviews; Product ranking / bestseller badge where available; Delivery offers; Bundles / free ink / freebies; Product positioning / key claims; Product URL; Date / time captured.
Epson	Same metrics as HP.
Canon	Same metrics as HP.
Brother	Same metrics as HP.

8. Keywords

The crawl should be focused on HP, Epson, Canon and Brother Ink Tank activity, while still allowing the engine to discover relevant product, promotional, social and e-commerce content.

General rule of thumb: Combine [Brand] + [Ink Tank terminology] + [commercial / product / comparison term]

Example keyword combinations: HP + Smart Tank + promotion; Epson + EcoTank + ราคา; Canon + MegaTank + รีวิว; Brother + Ink Tank + ขายแล้ว; HP + เครื่องพิมพ์แท็งก์หมึก + โปรโมชัน

Brand	English keyword examples	Thai equivalents / keyword examples
HP	HP Smart Tank; HP Ink Tank; HP Smart Tank Printer; HP Ink Tank Printer; HP Smart Tank all-in-one; refillable ink printer	HP Smart Tank; HP แท็งก์หมึก; HP แทงค์หมึก; HP เครื่องพิมพ์แท็งก์หมึก; HP ปรี้นเตอร์เติมหมึก; HP เครื่องพิมพ์เติมหมึก
Epson	Epson EcoTank; Epson Ink Tank; Epson EcoTank Printer; Epson Ink Tank Printer; Epson EcoTank all-in-one; refillable ink printer	Epson EcoTank; Epson แท็งก์หมึก; Epson แทงค์หมึก; Epson เครื่องพิมพ์แท็งก์หมึก; Epson ปรี้นเตอร์เติมหมึก; Epson เครื่องพิมพ์เติมหมึก
Canon	Canon MegaTank; Canon Ink Tank; Canon MegaTank Printer; Canon Ink Tank Printer; Canon PIXMA G; Canon PIXMA G-series; refillable ink printer	Canon MegaTank; Canon แท็งก์หมึก; Canon แทงค์หมึก; Canon เครื่องพิมพ์แท็งก์หมึก; Canon PIXMA G; Canon ปรี้นเตอร์เติมหมึก; Canon เครื่องพิมพ์เติมหมึก
Brother	Brother Ink Tank; Brother Ink Tank Printer; Brother Refill Tank; Brother InkBenefit; Brother InkBenefit Printer; Brother DCP-T series	Brother Ink Tank; Brother แท็งก์หมึก; Brother แทงค์หมึก; Brother เครื่องพิมพ์แท็งก์หมึก; Brother เครื่องพิมพ์เติมหมึก; Brother InkBenefit

9. Data Cuts

Please structure the collection around the following data cuts shown in the table below.

Parameters: HP vs Epson, Canon and Brother; cover the last 3 months / 90 days (~28 May–28 August).

Retain the date of each ad, post, product listing, promotion or review where possible.

Keep the source URL and underlying content / evidence wherever possible.

Ensure analytical outputs are in English.

Data cut	What to capture	Suggested format
1. Online visibility / SOV (Share of Voice)	Number of ads; unique creatives; social posts; engagement (likes, comments, shares, views where available); e-commerce listings / SKUs; sellers; official / authorised listings; promotional / featured listings; units sold or other sales indicators where visible.	Brand × Month × Channel × Platform. Keep underlying source data so we can calculate relative visibility where appropriate.
2. Advertising / creatives	Platform, date, product / SKU, creative format (video / static / carousel etc.), headline / hook, key message, CTA, product claims / benefits, promotion and target audience / use case where identifiable.	One row = one ad / creative, with content and URL. Also tag by brand × month × platform × creative theme / format.

Data cut	What to capture	Suggested format
3. Social media activity	Post date, platform, product / SKU, post type, content / theme, key message, reach, engagement (likes, comments, shares/reposts) and views where available, promotional content and URL. Focus on Ink Tank-related content.	One row = one relevant social post. Aggregate by Brand × Month × Platform × Theme.
4. E-commerce presence, pricing, promotions & traction	Capture SKU-level e-commerce activity and commercial signals, including brand, model / SKU, product title, platform / retailer, seller / authorised-seller status, current & original price, discount, stock availability, ratings / reviews, bestseller / ranking, delivery offers, promotions / sales push (e.g. vouchers, cashback, free shipping / ink / accessories, bundles, warranty, instalments), promotion dates, and where visible, units sold, product views, engagement and other observable traction signals. Also identify heavily promoted products / SKUs based on the intensity of listings, promotions, sellers, ads / social activity and other visible push indicators. Do not infer promotional causality from response signals.	One row = one SKU / listing observation. Aggregate by Brand × Month × Platform / Retailer × Product / SKU, with promotion / push and observable traction captured as associated attributes / indicators.
5. Consumer sentiment / recommendation — <i>Low priority: Dependent on crawlability and coverage of relevant review sites, forums and consumer discussion sources; drop if reliable access / sufficient coverage cannot be achieved</i>	Relevant reviews / comments / posts; positive / neutral / negative sentiment; recommendation signals; and reasons / audience or consumer conversation themes such as print quality, running cost, ink yield, refill experience, reliability, speed, connectivity, ease of use, value, maintenance, setup and price.	One row = one relevant review / comment. Capture Brand × Product × Platform / Source × Month × Sentiment × Theme, plus original Thai text, English translation / summary and URL.

10. Data Output

Please structure the collected data so that it can directly support the POC dashboard and RAG engine, while allowing every insight to be traced back to its original source.

Retain the underlying data at the most granular level possible, then organise it into consistent fields so we can compare HP vs Epson, Canon and Brother and track activity month-on-month.

10.1 Source-level data

Please retain one row per relevant observation — e.g. one ad, social post, product listing, promotion or consumer review.

Example:

Date	Brand	Channel	Platform	Activity Type	Product / SKU	Key Metric / Content	Value / Observation	URL
15-Aug-26	HP	E-commerce	Shopee	Product listing	Smart Tank 580	Current Price	฿5,490	URL
16-Aug-26	Epson	Paid Media	Meta	Ad / Creative	EcoTank L3250	Key Message	Low-cost printing	URL

Date	Brand	Channel	Platform	Activity Type	Product / SKU	Key Metric / Content	Value / Observation	URL
17-Aug-26	Canon	Social	Facebook	Post	PIXMA G3730	Engagement	1,245	URL
18-Aug-26	Brother	Consumer Review	Shopee	Review	DCP-T520W	Sentiment	Positive – good value	URL

Retain the original content / evidence wherever possible — e.g. Thai ad / post copy, creative, product information and review text — rather than only extracting the metric. Retain the source URL for each observation. Where relevant, retain the date / time captured as well as the original publication / listing date.

10.2 Structured / normalised data: Analytical dataset

Please create a structured and normalised analytical dataset with consistent fields that allows the data to be aggregated, compared and trended across HP, Epson, Canon and Brother.

The basic structure should support: *Brand × Month × Channel × Platform × Product / SKU × Metric*

Example:

Month	Brand	Channel	Platform	Product / SKU	Metric	Value
Jun-26	HP	E-commerce	Shopee	Smart Tank 580	Avg. Price (THB)	฿5,590
Jun-26	Epson	E-commerce	Shopee	EcoTank L3250	Avg. Price (THB)	฿5,190
Jun-26	Canon	Paid Media	Meta	PIXMA G3730	Unique Ads	14
Jun-26	Brother	Social	Facebook	DCP-T520W	Relevant Posts	9
Jul-26	HP	E-commerce	Shopee	Smart Tank 580	Avg. Price (THB)	฿5,490
Jul-26	Epson	E-commerce	Shopee	EcoTank L3250	Avg. Price (THB)	฿5,090

The structure above should clearly show month-on-month trends — for example, June vs July vs August; HP vs competitor comparisons; platform / channel comparisons; product / SKU-level comparisons; and 3-month / quarterly summaries.

For pricing, please capture all Thailand e-commerce prices in Thai Baht (THB/฿) and retain the original listed price where available.

10.3 Traceability back to source

Please ensure the analytical data retains a reference / link back to the underlying source record wherever possible.

This is particularly important for the RAG engine: When it generates an insight or answer, we should be able to trace it back to the underlying ad, post, product listing, review or other source evidence.

Example:

Month	Brand	Product	Metric	Value	Source / Evidence
Aug-26	HP	Smart Tank 580	Promotions	14	Source IDs / URLs
Aug-26	Epson	EcoTank L3250	Promotions	18	Source IDs / URLs
Aug-26	Canon	PIXMA G3730	Avg. Rating	4.7	Source IDs / URLs

Month	Brand	Product	Metric	Value	Source / Evidence
Aug-26	Brother	DCP-T520W	Positive Reviews	82%	Source IDs / URLs

In short, the dashboard should show the analysis and insights, while the underlying datasets should preserve the evidence and source references behind those insights.

11. RAG Intelligence / Q&A (Query Capability)

Build a RAG layer that allows users to ask natural-language questions across the collected advertising, social, e-commerce and consumer data, and receive concise, evidence-backed answers.

Core Q&A capabilities

The RAG engine should be able to answer questions such as:

- **Competitive:** How does HP compare with Epson, Canon and Brother?
- **Visibility:** Which brands / products are most visible, and how has visibility changed over the last 3 months?
- **Marketing / messaging:** What are competitors emphasising, and how does this differ from HP?
- **Promotions:** Which products are being most heavily promoted, and what promotional tactics are being used?
- **Pricing:** How do HP's prices compare with competitors across platforms and SKUs?
- **Product push:** Which products / SKUs are receiving the strongest overall push?
- **Consumer response:** What are the main positive and negative consumer themes?
- **Emerging signals:** What successes, issues, threats or opportunities for HP are emerging?
- **Trends / changes:** What has changed across June, July and August?
- **Implications:** Based on the available evidence, what should HP pay attention to or act on?

RAG output requirements

- Synthesise across sources and data cuts where relevant, rather than simply retrieve individual records.
- Surface evidence-backed successes, issues, emerging signals, competitive threats and opportunities for HP.
- Clearly distinguish observed facts from interpretation / hypotheses; do not infer causality without evidence.
- Where useful, generate a chart or table to support the answer.
- Provide source links / evidence for key insights so the user can trace the answer back to the underlying data.
- Where possible, indicate evidence strength / confidence based on the supporting evidence.

Suggested answer format: Answer → Evidence / supporting metrics → Implication for HP → Sources

Implementation: Do not create a separate “successes / issues / emerging signals” dataset. Generate these insights dynamically from the underlying structured and source-level data. This maintains the brief's requirement that insights remain traceable to the underlying source evidence.

12. Dashboard / UI & Visual Specifications

The POC should present the competitive intelligence in a simple, intuitive dashboard that allows David Chiu to quickly understand how HP compares with Epson, Canon and Brother, what each brand is doing, which products / promotions are being pushed, what consumers are saying, and what the key competitive implications are.

As this is a POC, please do not try to build every possible chart or expose every scraped data point. The dashboard should show enough high-quality examples to demonstrate the intelligence engine's value, rather than becoming a comprehensive data dashboard.

Prioritise clarity, useful comparisons and actionable insights over data volume.

12.1 Information hierarchy

Priority	Section	What it should answer
1 — Highest	Executive Summary / Key Insights	What is happening and what matters? 3–5 key competitive insights, opportunities, threats or notable changes.
2	Competitive Visibility	How active / visible is each brand? Paid media, social and e-commerce activity, with monthly trends where available.
3	Creative & Messaging	What is each brand saying? Key themes, claims, hooks, positioning and representative creative examples.
4	Promotions & Pricing	How are brands pushing sales? Discounts, vouchers, bundles / free ink, promotions and price points in THB/฿.
5	Product / SKU Push	Which products are being pushed? Most visible / promoted SKUs, pricing, sales indicators, ratings / reviews and relevant claims.
6	Consumer Sentiment	What are consumers saying? Positive / negative sentiment, ratings and recurring reasons / themes.
7	Competitive Signals / Implications for HP	So what does this mean for HP? Emerging opportunities, competitor strengths, issues, threats and potential implications.
8 — Supporting	Source / Evidence	Can I verify this? Links / evidence behind the insights, ideally accessible from the relevant insight or metric.

12.2 Suggested wireframe / visual direction

Use a single-page dashboard as the primary view, with clear sections and the ability to drill down where useful.

Below is an example to communicate the desired information hierarchy and user journey, rather than serve as a fixed design specification.

HEADER / FILTERS	
HP Ink Tank Competitive Intelligence — Thailand May–Aug 2026 Brand: All Month: All	
ASK INTELLIGENCE / RAG	
Natural-language Q&A across all collected data, with evidence-backed answers, relevant charts / tables and source links	
1. EXECUTIVE SUMMARY — WHAT MATTERS?	
3–5 key competitive insights, opportunities, threats or notable changes.	
2. COMPETITIVE VISIBILITY	
Monthly comparison of HP vs Epson vs Canon vs Brother across Paid Media / Advertising, Social and E-commerce.	
3. CREATIVE & MESSAGING Key themes, claims, hooks, positioning and representative creative examples.	4. PROMOTIONS & PRICING Current / average pricing, discounts, vouchers, bundles and representative promotion examples.
5. PRODUCT / SKU PUSH Top promoted products / SKUs and observable product-level activity.	6. CONSUMER SENTIMENT Positive / negative themes, ratings / reviews and key reasons where available.
7. COMPETITIVE SIGNALS / IMPLICATIONS FOR HP	
Strengths, opportunities, threats, issues and notable changes supported by evidence.	
8. EVIDENCE / SOURCES	
Links to supporting ads, posts, product listings, reviews and other source records.	

12.3 Visual principles

Use the wireframe primarily as a guide to information hierarchy and layout, rather than as a fixed design specification.

Keep the interface clean and uncluttered, and make HP vs Epson vs Canon vs Brother comparisons immediately obvious.

Prioritise 3-month trends and meaningful comparisons over isolated numbers.

Use a mix of simple charts, KPI cards, tables and representative examples, and avoid filling the page with every scraped data point.

Where possible, allow a user to click / drill down from an insight → metric → underlying evidence / source.

The “Ask Intelligence” / RAG capability should be clearly visible but should complement, rather than overwhelm, the dashboard. All dashboard text and generated insights should be in English, and e-commerce prices should be shown in **THB/฿**.

You may determine the detailed UI implementation, chart types and styling, provided the above hierarchy and core functionality are retained.